# Homework 4 – Regression Consult

When people make decisions, their choices are often swayed by how information is presented. This is often described as "the framing effect." One example of this, is how commercials describe new medications. As an example, Nereus (tradipitant), a new drug for motion-induced vomiting like sea-sickness, is marketed with a **positive frame** as reducing vomiting incidence from 44% to 18% (a reduction of 60%). This marketing makes it seem more effective than a **negative frame** that says just 18% of patients still experienced vomiting, even though both statements are true.

Xu (2024) aimed to evaluate whether older adults are more or less vulnerable to framing than younger adults. They collected a sample of 696 adults aged 18 to 71 and exposed them to three different scenarios

**Scenario 1:** Rate Drug Efficacy (1 = very bad, 6 = very good)

- **Positive Frame:** 70 out of 100 patients felt better
- **Negative Frame:** 30 out of 100 patients did not feel better

**Scenario 2:** Likelihood to stop drinking cocoa (1 = surely stop, 6 = surely continue)

- **Positive Frame:** Stopping drinking hot cocoa reduces cadmium and cancer risk.
- **Negative Frame:** Continuing to drink hot cocoa increases cadmium and cancer risk.

**Scenario 3:** Preference for chance choice (1 = surely safe option, 6 = surely risky)

- **Positive Frame:** Choice between 200 saved vs 1/3 chance all 600 are saved.
- **Negative Frame:** Choice between 400 die vs 2/3 chance all 600 die.

In each scenario, participants were randomly assigned a positive or negative frame. They rated their response on a 1-6 scale. The researchers want to build three models, one for each scenario adjusting for simultaneous linear changes in demographic variables (gender, ethnicity, education level, political affiliation).

They want to evaluate whether, as adults age, the way choices are worded have more influence on their judgments and ultimate choices. If there is a change, are those changes driven by positive or negative framings? At what age does this effect onset?

**Your Goals:**

- Determine the correct analysis strategy and conduct the necessary analyses.
  - Hint 1: Ensure to assess and communicate results about the model fit.
  - Hint 2: Ensure to think about whether or which interactions need to be included.
  - Hint 3: You will need run various post hoc procedures (e.g., `emmeans`, `emtrends` or `johnson_neyman`).
  - Hint 4: Looking at the paper may be helpful. They do include `R` code, but I expect you to do a much better job using the techniques we discussed (they do somethings differently)
- Write a clear, concise, complete, and correct report conveying an answer to their research question.

The data are in `HW4Data.csv` but they will require some manual and `tidyverse` cleaning.

# 1 Loading and Cleaning the Data

```r
# Load Data
dat <- read_csv("HW4data.csv",
                col_names = F,
                col_types = cols(.default = col_character()))

# Clean Data
dat <- dat |>
  slice(1:698) |>      # rows 699-844 are empty rows
  select(X1:X32)       # X33-X52 are empty columns

names(dat) <- str_trim(as.character(dat[2, ]))  # row 2 becomes the column names

dat <- dat |>
  slice(-(1:2)) |>    # remove the question-text row and the names row
  rename(duration = duratin) |>
  type_convert(col_types = cols()) |>
  select(-c(confirmation, Q_RecaptchaScore, userLanguage, distributionChannel, startDate, endDate))

# Clean age attention check
dat <- dat |>
  mutate(band   = stri_trans_nfkc(ageattentioncheck),   # unbolding
         age_ok = is.na(band) |                         # early respondents weren't asked
                  between(age, parse_number(band), parse_number(str_remove(band, ".*-"))))

dat <- dat |>
  filter(age_ok, is.na(attentioncheck) | attentioncheck == 1) |>
  select(-c(ageattentioncheck, attentioncheck, band, age_ok))

# Remove "prefer not to say" answers / fix wordings
framing <- dat |>
  mutate(across(c(gender, ethnicity, politics, education), ~ na_if(.x, "Prefer not to say"))) |>
  drop_na(gender, ethnicity, politics, education) |>
  transmute(
    id = responseID,
    age,
    gender    = fct_infreq(gender),       # most common level is reference
    ethnicity = fct_infreq(ethnicity),
    politics  = fct_infreq(politics),
    education = factor(education,
                       levels = c("Less than high school", "High school graduate",
                                  "Some college", "2 year degree", "4 year degree",
                                  "Professional degree", "Doctorate")) |>
                fct_relevel("4 year degree"),   # largest group is reference
    frame_attribute   = factor(if_else(treatment1 == "pos1", "Positive", "Negative"),
                           levels = c("Negative", "Positive")),
    frame_goal  = factor(if_else(treatment2 == "pos2", "Positive", "Negative"),
                           levels = c("Negative", "Positive")),
    frame_risky = factor(if_else(treatment3 == "pos3", "Positive", "Negative"),
                           levels = c("Negative", "Positive")),
    resp_attribute   = pick1,
    resp_goal  = pick2,
    resp_risky = pick3
  )

```

```r
# for plotting later
framing_long <- framing |>
  pivot_longer(c(starts_with("frame_"), starts_with("resp_")),
               names_to = c(".value", "scenario"), names_sep = "_") |>
  mutate(scenario = factor(scenario,
                           levels = c("attribute", "goal", "risky"),
                           labels = c("Attribute (drug)",
                                      "Goal (cocoa)",
                                      "Risky choice (disease)")))
```

When I loaded the .csv file, there were extra columns and rows that were empty and needed to be removed. Additionally the first two rows were the full question followed by a shortened name for the column, so I made the shortened name the column header and removed the row containing the survey question, and removed columns that wouldn't be useful to our analysis, including reCAPTCHA score, confirmation, language, distribution style, and start and end date. Next, like the paper, I removed participants who didn't pass the attention and age attention checks, then removed the columns. Lastly, I chose to remove patients who answered "prefer not to say" for a question (n = 10) and fixed some of the wordings of the framing columns so it was easier for me to understand.

## 2 Exploratory Data Analyses

```r
framing_long |>
  group_by(scenario, frame) |>
  summarize(n        = sum(!is.na(resp)),
            mean     = mean(resp, na.rm = TRUE),
            sd       = sd(resp, na.rm = TRUE),
            mean_age = mean(age),
            .groups  = "drop") |>
  kable(digits = 2,
        col.names = c("Scenario", "Frame", "n", "Mean rating", "SD", "Mean age"))
```

Table 1: Ratings (1–6) and participant age by scenario and assigned frame.

| Scenario | Frame | n | Mean rating | SD | Mean age |
|---|---|---|---|---|---|
| Attribute (drug) | Negative | 358 | 3.84 | 0.80 | 40.72 |
| Attribute (drug) | Positive | 320 | 4.49 | 0.62 | 41.08 |
| Goal (cocoa) | Negative | 329 | 2.00 | 1.09 | 40.88 |
| Goal (cocoa) | Positive | 349 | 2.52 | 1.40 | 40.89 |
| Risky choice (disease) | Negative | 337 | 3.59 | 1.50 | 41.90 |
| Risky choice (disease) | Positive | 340 | 2.89 | 1.38 | 39.88 |

Random assignment worked as intended. In each scenario, the two frames have similar group sizes and similar mean ages (Table 1).

**Attribute:** The drug was rated as more effective under the positive frame than under the negative frame.

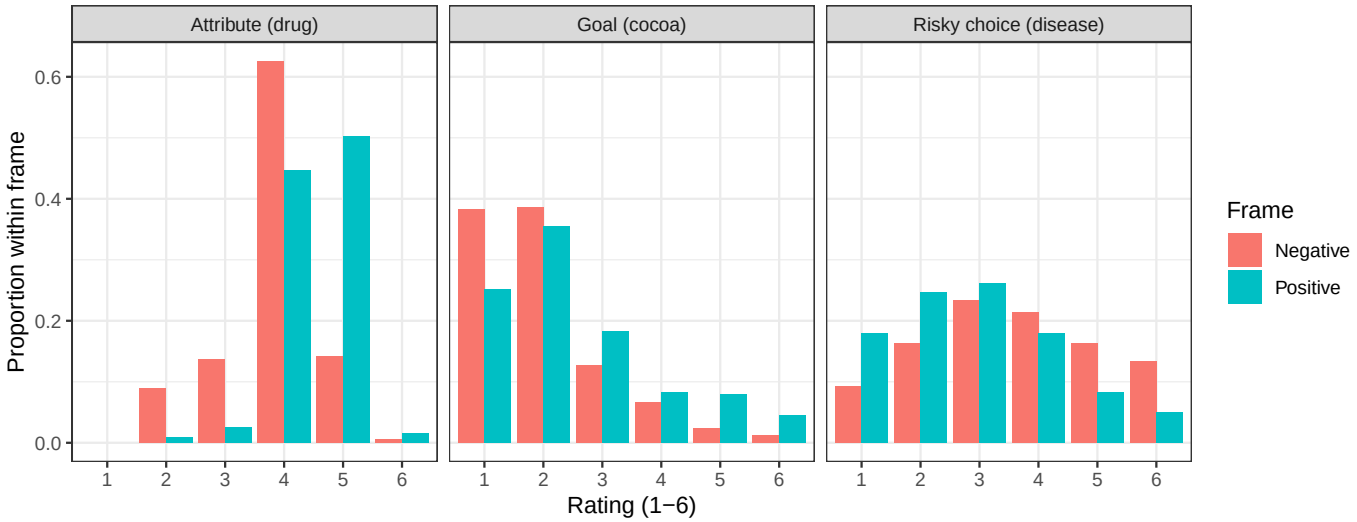
**Goal:** Under the negative frame, participants were more likely to say they would stop drinking cocoa.

**Risky choice:** The negative frame pushed participants toward the risky option.

```r
framing_long |>
  drop_na(resp) |>
  ggplot(aes(x = factor(resp), fill = frame)) +
  geom_bar(aes(y = after_stat(prop), group = frame), position = "dodge") +
```

```
  facet_wrap(~ scenario) +
  labs(x = "Rating (1-6)", y = "Proportion within frame", fill = "Frame") +
  theme_bw()
```

Figure 1: Distribution of ratings by frame within each scenario.



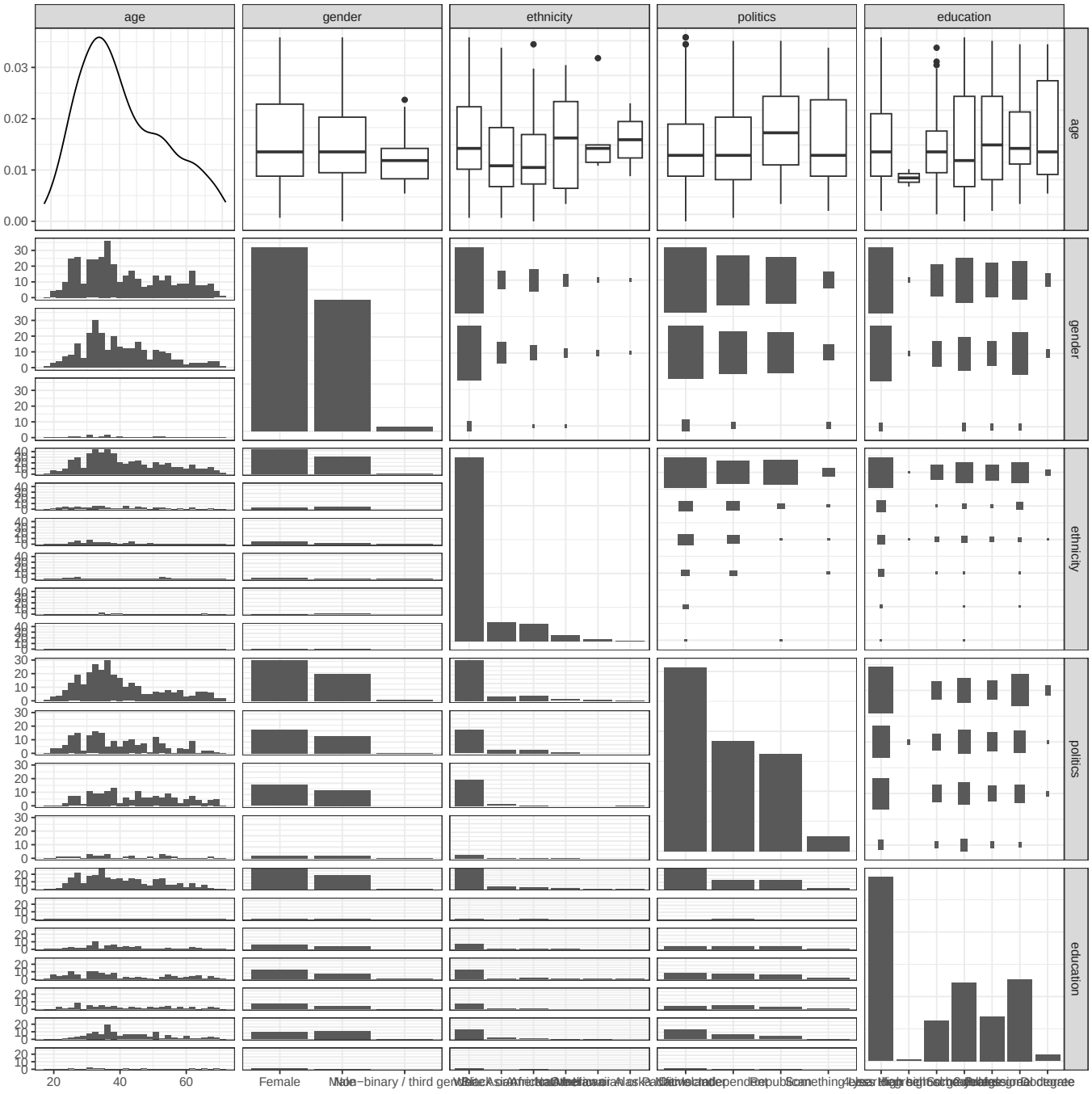

The responses are on a discrete, bounded 1–6 scale (Figure 1). In the attribute scenario, almost all ratings are 4 or 5. In the goal scenario, ratings pile up at 1–2, close to the floor of the scale. As a result, the residuals will fall in bands, and their distribution may not be Gaussian in the tails. We check this in the residual analysis for each model.

### 2.0.1 Pairwise plots

All three models share the same adjustment variables, so we check them once for strong relationships with each other.

```
ggpairs(framing,
        columns  = c("age", "gender", "ethnicity", "politics", "education"),
        progress = FALSE) +
  theme_bw()
```

Figure 2: Pairwise plots of age and the demographic adjustment variables.




None of the demographic variables is strongly related to another (Figure 2). The clearest pattern is that Republicans are a few years older on average than the other political groups. That relationship is mild, and we confirm that it does not inflate the standard errors with variance inflation factors after fitting the models.

```
ggpairs(framing, columns = c("resp_attribute", "frame_attribute", "age"),
        mapping = aes(color = frame_attribute, alpha = 0.5), progress = FALSE) +
  theme_bw()

ggpairs(framing, columns = c("resp_goal", "frame_goal", "age"),
        mapping = aes(color = frame_goal, alpha = 0.5), progress = FALSE) +
  theme_bw()
```

```r

ggpairs(framing, columns = c("resp_risky", "frame_risky", "age"),
        mapping = aes(color = frame_risky, alpha = 0.5), progress = FALSE) +
  theme_bw()
```

### 2.0.2 Does the framing effect change with age?

```r
framing_long |>
  drop_na(resp) |>
  ggplot(aes(x = age, y = resp, color = frame, fill = frame)) +
  geom_jitter(width = 0, height = 0.15, alpha = 0.25, size = 0.8) +
  geom_smooth(method = "lm", formula = y ~ x) +
  geom_smooth(method = "loess", formula = y ~ x, se = FALSE, linetype = "dashed", linewidth = 0.6) +
  facet_wrap(~ scenario) +
  scale_y_continuous(breaks = 1:6) +
  labs(x = "Age (years)", y = "Rating (1-6)", color = "Frame", fill = "Frame") +
  theme_bw()
```

Figure 4: Ratings against age by frame. Solid lines are linear fits with 95% confidence bands, and dashed lines are LOESS smooths. Points are jittered vertically.

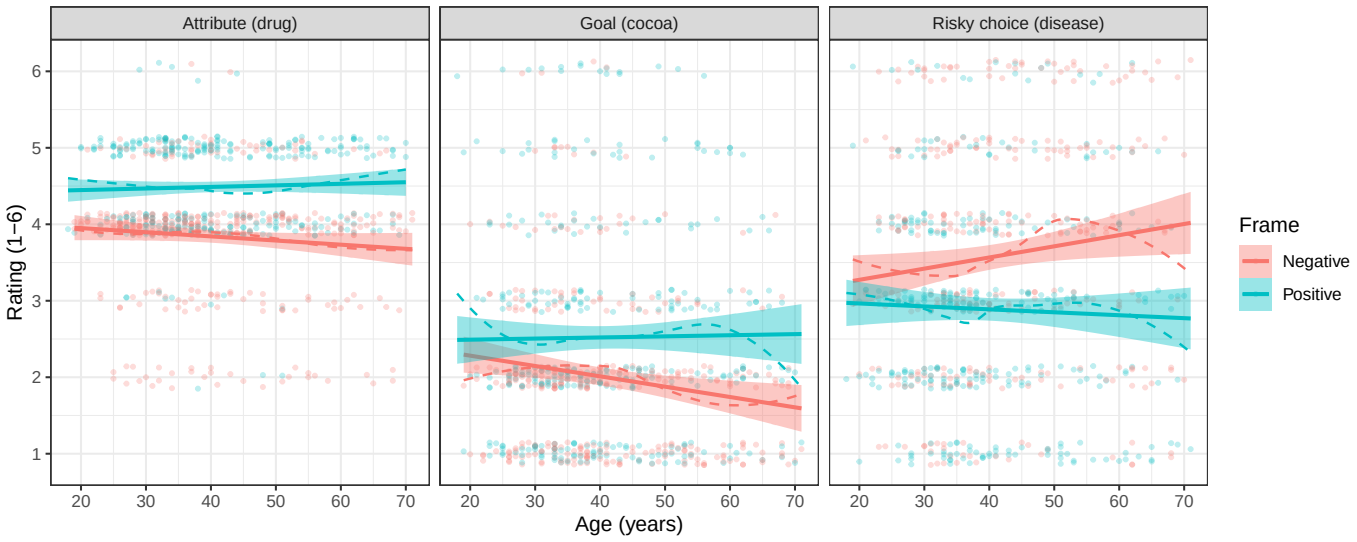



Figure 4 previews the research question. In every scenario, the positive-frame line is nearly flat across age, while the negative-frame line moves away from it as age increases:

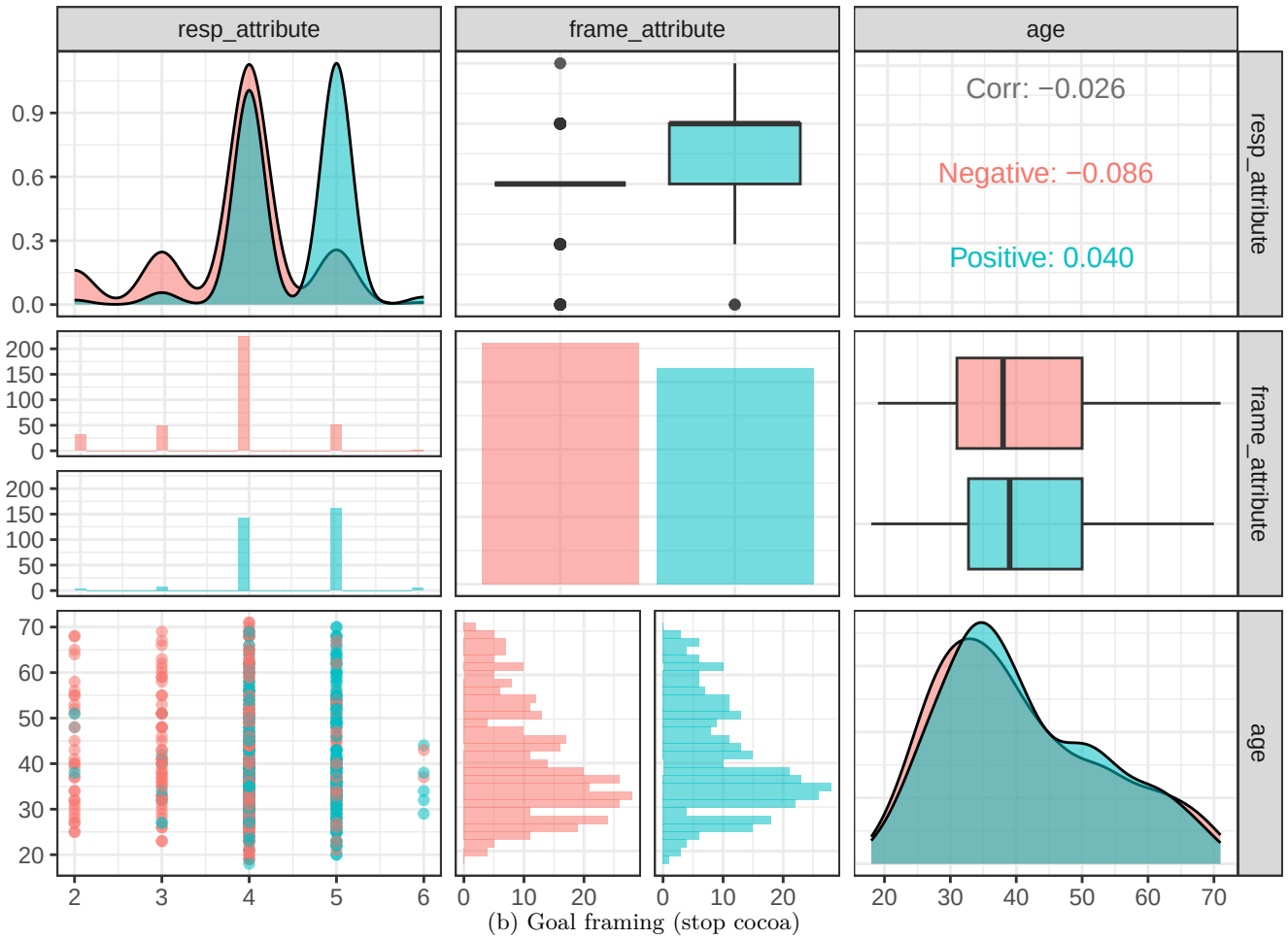
- **Attribute:** Under the negative frame, older participants rate the drug lower.
- **Goal:** Under the negative frame, older participants are more likely to say they would stop.
- **Risky choice:** Under the negative frame, older participants lean more toward the risky option.

In all three cases, the gap between the frames, which is the framing effect, appears to widen with age. The change seems to come mainly from the negative frame. This pattern suggests a frame × age interaction in each model, which we test formally in the regression analyses.

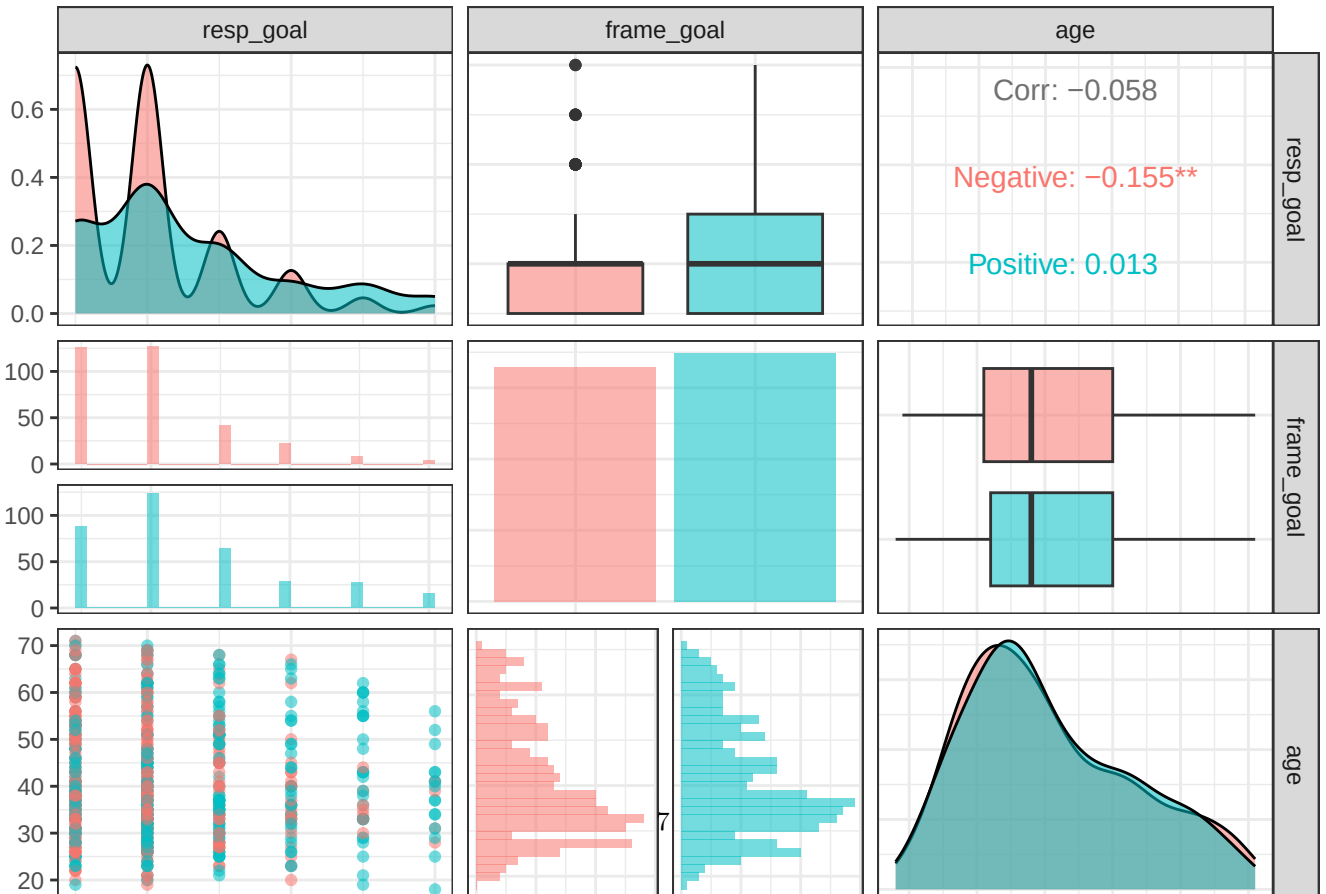
The relationships are weak: the within-frame correlations between age and rating are all below 0.2 in absolute value. The LOESS curves stay close to the straight lines, so a linear age term looks adequate. The curves wander more at the oldest ages, where there are few participants.

Figure 3: Pairwise plots of the rating, frame, and age for each scenario.

(a) Attribute framing (drug efficacy)




(b) Goal framing (stop cocoa)



# 3 Regression Analyses

```r
library(broom)          # tidy(), augment()
library(car)            # vif()
library(effectsize)     # F_to_omega2(), omega_squared(), interpret_omega_squared(), standardize_parameters(
library(emmeans)        # emmeans(), emtrends(), pairs()
library(interactions)   # johnson_neyman()
library(sandwich)       # vcovHC() for the robustness check
```

**Analysis strategy.** We fit one multiple linear regression per scenario:

$$\widehat{\text{Rating}}_i = \beta_0 + \underbrace{\cdots}_{\text{gender, ethnicity, politics, education}} + \beta_1 I(\text{Frame}_i = \text{Pos}) + \beta_2\, \text{Age}_i + \beta_3\, I(\text{Frame}_i = \text{Pos})\, \text{Age}_i$$

- **Why the frame × age interaction:** The research question is whether the *effect of the frame* changes with age, which is exactly what the interaction measures. $\beta_3$ is the difference between the age slopes of the positive and negative frames. The paper fit the same structure.
- **Term order:** The demographic adjustment variables go first, so the sequential (Type I) `anova()` tests for frame, age and the interaction are all adjusted for demographics. The interaction is entered last, so its $F$-test matches its $t$-test.
- **Age is left uncentered:** As a result, $\beta_1$ is the framing effect at age 0, which is not meaningful by itself. We interpret the frame and age effects through estimated marginal means and trends instead.

Each model is checked against the conditions for use:

- *Representativeness:* the sample is an MTurk convenience sample, which we discuss in the Discussion.
- *Independence:* each person contributes one rating per model, and frames were randomly assigned.
- *Linearity:* assessed with the LOESS curves in Figure 4.
- *Gaussian, constant-variance errors:* assessed with residual plots.
- *Influence:* assessed with leverage, Cook's distance, DFFITS and studentized residuals.
- *Multicollinearity:* assessed with VIFs.

Post hoc procedures then answer the three research questions:

1. **Does the framing effect change with age?** We test the interaction with an $F$-test and give its effect size.
2. **Is any change driven by the positive or the negative frame?** We use `emtrends()` to estimate the age slope within each frame.
3. **At what age does the effect onset?** We estimate the framing effect at specific ages with `emmeans()`, and we run a Johnson-Neyman analysis with false-discovery-rate control.

```r
# Residual plots: Q-Q plot and residuals vs fitted values (+/- 2 and 3 sigma)
resid_plots <- function(mod) {
  d <- augment(mod)
  s <- sigma(mod)
  qq <- ggplot(d, aes(sample = .resid)) +
    stat_qq(alpha = 0.4) + stat_qq_line() +
    labs(x = "Theoretical Quantiles", y = "Observed Quantiles") +
    theme_bw()
  rf <- ggplot(d, aes(.fitted, .resid)) +
    geom_point(alpha = 0.4) +
    geom_hline(yintercept = 0, linetype = "dashed", color = "deeppink3") +
    geom_hline(yintercept = c(-2, 2) * s, linetype = "dotted", color = "steelblue") +
    geom_hline(yintercept = c(-3, 3) * s, linetype = "dotted", color = "darkorange") +
```

```r
    labs(x = expression("Fitted Values (" * hat(Y) * ")"), y = "Residual") +
    theme_bw()
  list(qq = qq, rf = rf)
}

# Influence plots: leverage, Cook's distance, DFFITS, studentized residuals
influence_plots <- function(mod) {
  n <- nobs(mod); p <- length(coef(mod))
  d <- tibble(obs = seq_len(n),
              Leverage                = hatvalues(mod),
              `Cook's Distance`       = cooks.distance(mod),
              DFFITS                  = dffits(mod),
              `Studentized Residual`  = rstudent(mod)) |>
    pivot_longer(-obs, names_to = "measure") |>
    mutate(measure = fct_inorder(measure))
  cuts <- bind_rows(
    tibble(measure = "Leverage",             y = c(2, 3) * p / n,           level = c("Moderate", "High")),
    tibble(measure = "Cook's Distance",      y = qf(c(0.2, 0.5), p, n - p), level = c("Moderate", "High")),
    tibble(measure = "DFFITS",               y = c(-2, 2, -3, 3) * sqrt(p / n),
           level = rep(c("Moderate", "High"), each = 2)),
    tibble(measure = "Studentized Residual", y = c(-2, 2, -3, 3),
           level = rep(c("Moderate", "High"), each = 2))
  ) |> mutate(measure = factor(measure, levels = levels(d$measure)))
  ggplot(d, aes(obs, value)) +
    geom_point(size = 0.6, alpha = 0.6) +
    geom_hline(data = cuts, aes(yintercept = y, color = level), linetype = "dotted") +
    facet_wrap(~ measure, scales = "free_y") +
    scale_color_manual(values = c(Moderate = "steelblue", High = "darkorange")) +
    labs(x = "Observation Number", y = NULL, color = "Cutoff") +
    theme_bw()
}

# Coefficient table with standardized coefficients
coef_table <- function(mod) {
  tidy(mod) |>
    mutate(std = standardize_parameters(mod)$Std_Coefficient,
           p.value = if_else(p.value < 0.0001, "<0.0001", as.character(round(p.value, 4)))) |>
    kable(digits = 3, col.names = c("Term", "Estimate", "SE", "t", "p", "Std-Coef"))
}

# Partial omega-squared for each term in the (Type I) ANOVA
term_effects <- function(mod) {
  omega_squared(anova(mod)) |>
    as_tibble() |>
    mutate(Size = as.character(interpret_omega_squared(Omega2_partial, rules = "cohen1992"))) |>
    select(Parameter, Omega2_partial, CI_low, CI_high, Size) |>
    kable(digits = 3, col.names = c("Term", "Partial omega^2", "Lower", "Upper", "Size"))
}

# Estimated marginal means across age, by frame
emm_plot <- function(mod, frame, ylab) {
  emmeans(mod, specs = frame, by = "age", at = list(age = 18:71)) |>
    as.data.frame() |>
    ggplot(aes(age, emmean, color = .data[[frame]], fill = .data[[frame]])) +
    geom_ribbon(aes(ymin = lower.CL, ymax = upper.CL), alpha = 0.2, color = NA) +
    geom_line(linewidth = 1) +
```

```
    labs(x = "Age (years)", y = ylab, color = "Frame", fill = "Frame") +
    theme_bw()
}

# Key numbers for inline text
key_stats <- function(mod, frame) {
  s   <- summary(mod)
  int <- tidy(mod) |> filter(str_detect(term, ":age"))
  emt <- emtrends(mod, specs = frame, var = "age") |> test() |> as.data.frame()
  list(n = nobs(mod), r2 = s$r.squared, adj = s$adj.r.squared,
       F = s$fstatistic[1], df1 = s$fstatistic[2], df2 = s$fstatistic[3],
       Fp = pf(s$fstatistic[1], s$fstatistic[2], s$fstatistic[3], lower.tail = FALSE),
       int_b = int$estimate, int_p = int$p.value,
       neg_b = emt$age.trend[1], neg_p = emt$p.value[1],
       pos_b = emt$age.trend[2], pos_p = emt$p.value[2])
}
fp <- function(p) if (p < 0.001) "p < 0.001" else paste0("p = ", formatC(p, format = "f", digits = 3))
```

## 3.1 Rating Drug Efficacy

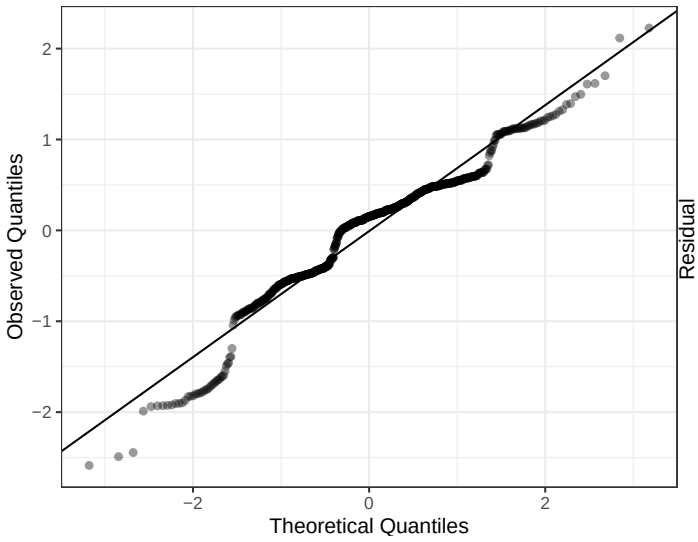
```
m_att <- lm(resp_attribute ~ gender + ethnicity + politics + education +
              frame_attribute * age,
            data = framing)
k_att <- key_stats(m_att, "frame_attribute")
```

**Conditions.**

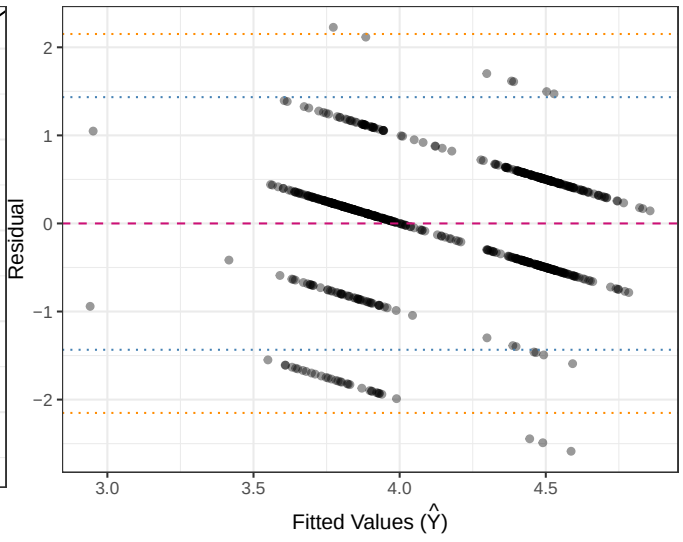
```
r <- resid_plots(m_att); r$qq; r$rf
```

Figure 5: Residual diagnostics for the attribute-framing model.

(a) Q-Q plot

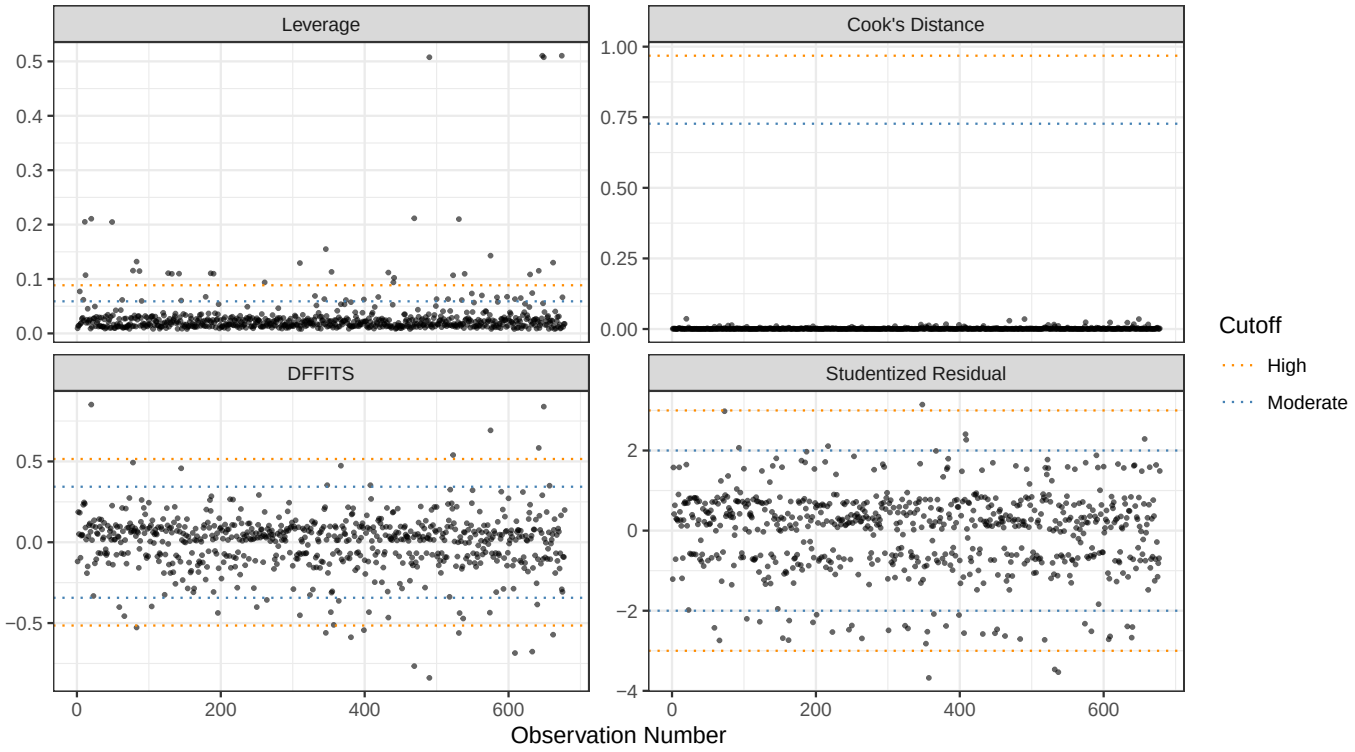



(b) Residuals vs. fitted values




```
influence_plots(m_att)
```

Figure 6: Influence diagnostics for the attribute-framing model.




```r
# Residual spread by frame (constant variance)
augment(m_att) |> group_by(frame_attribute) |> summarize(resid_sd = sd(.resid)) |> kable(digits = 3)
```

| frame_attribute | resid_sd |
|---|---|
| Negative | 0.778 |
| Positive | 0.619 |

```r
# VIF on the additive model (the interaction inflates VIF by construction)
vif(lm(resp_attribute ~ gender + ethnicity + politics + education + frame_attribute + age,
       data = framing))
```

```
                    GVIF Df GVIF^(1/(2*Df))
gender          1.069444  2        1.016926
ethnicity       1.151129  5        1.014174
politics        1.181429  3        1.028177
education       1.173558  6        1.013426
frame_attribute 1.030213  1        1.014994
age             1.059931  1        1.029530
```

```r
# Box-Cox: lambda that maximizes the likelihood
bc <- MASS::boxcox(m_att, plotit = FALSE); bc$x[which.max(bc$y)]
```

```
[1] 2
```

- **Gaussian errors:** Because the response is a 1–6 rating with nearly all values at 4 or 5, the residuals fall in diagonal bands (Figure 5). The Q-Q plot shows a heavier lower tail, i.e., left skew. With n = 678, the coefficient tests are robust to this, because the sampling distributions of the $\hat{\beta}$s are approximately Gaussian by the CLT.

- **Constant variance:** The residuals are somewhat more spread out under the negative frame than the positive frame. Below, we check whether this matters using heteroscedasticity-robust standard errors.
- **Box-Cox:** It pushes toward $\lambda = 2$, which reflects the pile-up near the top of the scale. Squaring a Likert rating has no sensible interpretation, so we keep the response untransformed.
- **Influence:** The few high-leverage points (Figure 6) are participants in demographic categories with only two people. None of them is influential: every Cook's distance is far below the moderate cutoff, and the studentized residuals are mostly within $\pm 3$.
- **Multicollinearity:** All $\text{GVIF}^{1/(2\,\text{Df})}$ values are close to 1, so there is no multicollinearity.

**Overall $F$-test and effect size.**

```
anova(m_att) |> as.data.frame() |> kable(digits = 3)
```

| | Df | Sum Sq | Mean Sq | F value | Pr(>F) |
|---|---|---|---|---|---|
| gender | 2 | 0.305 | 0.153 | 0.297 | 0.743 |
| ethnicity | 5 | 5.881 | 1.176 | 2.288 | 0.045 |
| politics | 3 | 0.484 | 0.161 | 0.314 | 0.815 |
| education | 6 | 4.680 | 0.780 | 1.517 | 0.170 |
| frame_attribute | 1 | 71.293 | 71.293 | 138.655 | 0.000 |
| age | 1 | 0.155 | 0.155 | 0.302 | 0.583 |
| frame_attribute:age | 1 | 1.414 | 1.414 | 2.751 | 0.098 |
| Residuals | 658 | 338.330 | 0.514 | NA | NA |

```
F_to_omega2(f        = k_att$F,
            df       = k_att$df1,
            df_error = k_att$df2)
```

```
Omega2 (partial) |       95% CI
------------------------------
0.18             | [0.12, 1.00]

- One-sided CIs: upper bound fixed at [1.00].
```

```
interpret_omega_squared(F_to_omega2(k_att$F, k_att$df1, k_att$df2)$Omega2_partial, rules = "cohen1992")
```

```
[1] "medium"
(Rules: cohen1992)
```

The model accounts for a statistically discernible share of the variation in efficacy ratings ($R^2 = 0.199$, adjusted $R^2 = 0.176$; $F(19, 658) = 8.62$, $p < 0.001$). Most of this comes from the frame (Table 2).

```
term_effects(m_att)
```

Table 2: Effect size (partial omega-squared) for each term, attribute framing.

| Term | Partial omega^2 | Lower | Upper | Size |
|---|---|---|---|---|
| gender | 0.000 | 0.000 | 1 | very small |
| ethnicity | 0.009 | 0.000 | 1 | very small |
| politics | 0.000 | 0.000 | 1 | very small |
| education | 0.005 | 0.000 | 1 | very small |
| frame_attribute | 0.169 | 0.128 | 1 | medium |

Table 2: Effect size (partial omega-squared) for each term, attribute framing.

| Term | Partial omega^2 | Lower | Upper | Size |
|---|---|---|---|---|
| age | 0.000 | 0.000 | 1 | very small |
| frame_attribute:age | 0.003 | 0.000 | 1 | very small |

```
coef_table(m_att)
```

Table 3: Coefficients for the attribute-framing model.

| Term | Estimate | SE | t | p | Std-Coef |
|---|---|---|---|---|---|
| (Intercept) | 4.053 | 0.140 | 28.953 | <0.0001 | -0.354 |
| genderMale | -0.007 | 0.057 | -0.114 | 0.909 | -0.008 |
| genderNon-binary / third gender | 0.105 | 0.233 | 0.449 | 0.6533 | 0.133 |
| ethnicityAsian | -0.010 | 0.102 | -0.094 | 0.9253 | -0.012 |
| ethnicityBlack or African American | 0.198 | 0.107 | 1.849 | 0.0649 | 0.250 |
| ethnicityOther | -0.106 | 0.171 | -0.620 | 0.5357 | -0.134 |
| ethnicityAmerican Indian or Alaska Native | -0.809 | 0.325 | -2.490 | 0.013 | -1.024 |
| ethnicityNative Hawaiian or Pacific Islander | 0.305 | 0.510 | 0.599 | 0.5496 | 0.386 |
| politicsIndependent | 0.043 | 0.068 | 0.630 | 0.529 | 0.054 |
| politicsRepublican | -0.077 | 0.073 | -1.061 | 0.2892 | -0.098 |
| politicsSomething else | -0.099 | 0.152 | -0.650 | 0.5161 | -0.125 |
| educationLess than high school | -0.553 | 0.515 | -1.074 | 0.2834 | -0.699 |
| educationHigh school graduate | -0.134 | 0.101 | -1.325 | 0.1855 | -0.170 |
| educationSome college | 0.003 | 0.079 | 0.044 | 0.9651 | 0.004 |
| education2 year degree | 0.059 | 0.098 | 0.605 | 0.5452 | 0.075 |
| educationProfessional degree | -0.109 | 0.078 | -1.399 | 0.1622 | -0.137 |
| educationDoctorate | 0.053 | 0.232 | 0.226 | 0.8209 | 0.067 |
| frame_attributePositive | 0.352 | 0.194 | 1.817 | 0.0697 | 0.834 |
| age | -0.005 | 0.003 | -1.506 | 0.1327 | -0.072 |
| frame_attributePositive:age | 0.008 | 0.005 | 1.658 | 0.0977 | 0.118 |

The frame × age interaction is not statistically discernible ($\hat{\beta}_3 = 0.0075$, p = 0.098), and its effect size is very small. We still run the post hoc procedures, because the research question asks about them directly. They should be read as describing trends rather than confirming a change with age.

**Post hoc testing.**

```
# Age slope within each frame: which frame drives any change?
emt_att <- emtrends(m_att, specs = "frame_attribute", var = "age")
test(emt_att) |> as.data.frame() |> kable(digits = 4)
```

| frame_attribute | age.trend | SE | df | t.ratio | p.value |
|---|---|---|---|---|---|
| Negative | -0.0046 | 0.0030 | 658 | -1.5055 | 0.1327 |
| Positive | 0.0029 | 0.0034 | 658 | 0.8632 | 0.3883 |

```
# Difference in slopes (= the interaction coefficient)
pairs(emt_att, reverse = TRUE) |> as.data.frame() |> kable(digits = 4)
```

| contrast | estimate | SE | df | t.ratio | p.value |
|---|---|---|---|---|---|
| Positive - Negative | 0.0075 | 0.0045 | 658 | 1.6585 | 0.0977 |

```r
# Framing effect (Positive - Negative) at selected ages, with Cohen's d
emm_att <- emmeans(m_att, specs = "frame_attribute", by = "age",
                   at = list(age = c(20, 30, 40, 50, 60, 70)))
pairs(emm_att, reverse = TRUE) |>
  as.data.frame() |>
  mutate(d = estimate / sd(framing$resp_attribute)) |>
  kable(digits = 3)
```

| contrast | age | estimate | SE | df | t.ratio | p.value | d |
|---|---|---|---|---|---|---|---|
| Positive - Negative | 20 | 0.502 | 0.110 | 658 | 4.565 | 0 | 0.636 |
| Positive - Negative | 30 | 0.577 | 0.075 | 658 | 7.737 | 0 | 0.731 |
| Positive - Negative | 40 | 0.652 | 0.056 | 658 | 11.623 | 0 | 0.826 |
| Positive - Negative | 50 | 0.728 | 0.070 | 658 | 10.454 | 0 | 0.921 |
| Positive - Negative | 60 | 0.803 | 0.103 | 658 | 7.780 | 0 | 1.016 |
| Positive - Negative | 70 | 0.878 | 0.143 | 658 | 6.123 | 0 | 1.111 |

```r
emm_plot <- function(mod, frame, ylab) {
  emmeans(mod, specs = frame, by = "age", at = list(age = 18:71),
          nuisance = c("gender", "ethnicity", "politics", "education")) |>
    as.data.frame() |>
    ggplot(aes(age, emmean, color = .data[[frame]], fill = .data[[frame]])) +
    geom_ribbon(aes(ymin = lower.CL, ymax = upper.CL), alpha = 0.2, color = NA) +
    geom_line(linewidth = 1) +
    labs(x = "Age (years)", y = ylab, color = "Frame", fill = "Frame") +
    theme_bw()
}
```

```r
# Same model with the frame coded 0/1 so johnson_neyman() can treat it as the focal predictor
m_att_jn <- lm(resp_attribute ~ gender + ethnicity + politics + education + pos * age,
               data = framing |> mutate(pos = as.numeric(frame_attribute == "Positive")))
jn_att <- johnson_neyman(m_att_jn, pred = pos, modx = age, control.fdr = TRUE)
jn_att
```

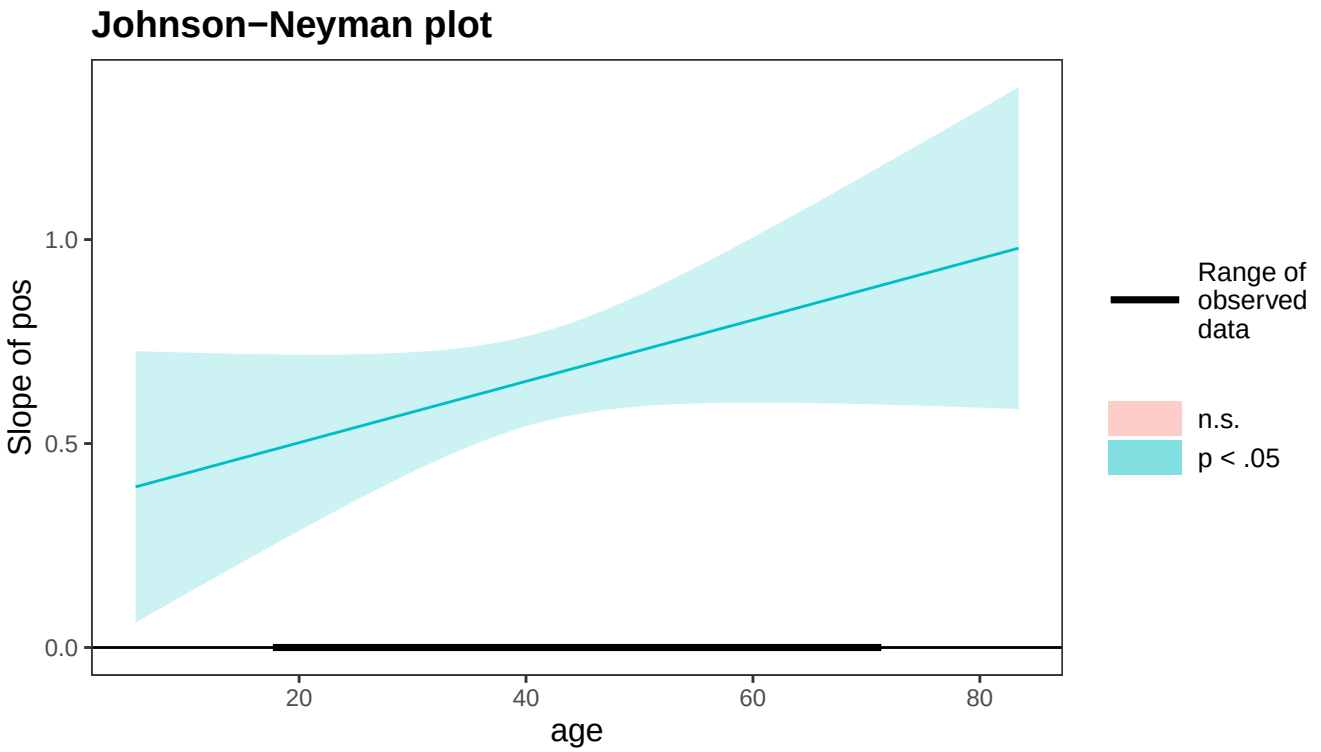
```
JOHNSON-NEYMAN INTERVAL

When age is INSIDE the interval [1.77, 516.35], the slope of pos is p <
.05.

Note: The range of observed values of age is [18.00, 71.00]

Interval calculated using false discovery rate adjusted t = 1.96
```

Figure 7: Johnson-Neyman plot: the framing effect (positive minus negative) across age, attribute framing.




**Summary.**

- **Framing effect:** The positive frame ("70 felt better") produces higher efficacy ratings than the negative frame ("30 did not feel better") at every age from 18 to 71 (Figure 7). The effect is therefore present across the whole observed age range rather than "onsetting" at a particular age.
- **Change with age:** The estimated gap grows by about 0.08 points per decade, but this increase is not statistically discernible ($p = 0.098$).
- **Direction:** The negative-frame slope (-0.046 per decade, $p = 0.133$) and the positive-frame slope (0.029 per decade, $p = 0.388$) are both indistinguishable from zero. Descriptively, though, the widening comes from older adults rating the drug *lower* under the negative frame.

## 3.2 Likelihood to Stop Drinking Cocoa

```
m_goal <- lm(resp_goal ~ gender + ethnicity + politics + education +
               frame_goal * age,
             data = framing)
k_goal <- key_stats(m_goal, "frame_goal")
```

**Conditions.**

```
r <- resid_plots(m_goal); r$qq; r$rf
```

```
influence_plots(m_goal)
```

Figure 8: Residual diagnostics for the goal-framing model.

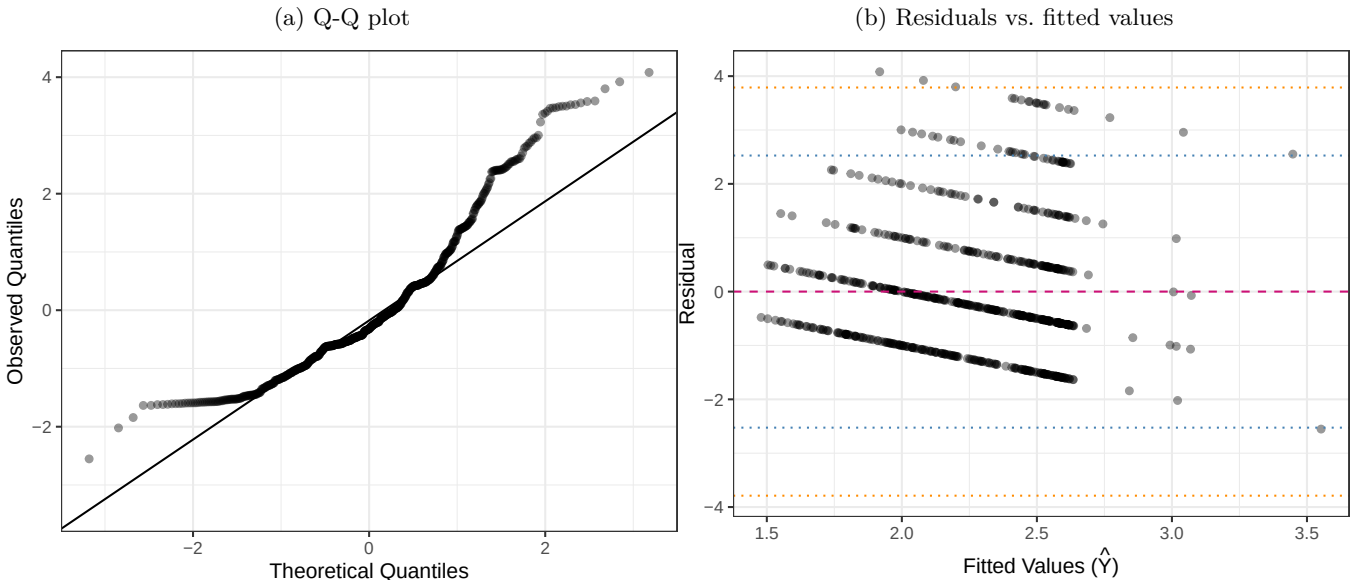
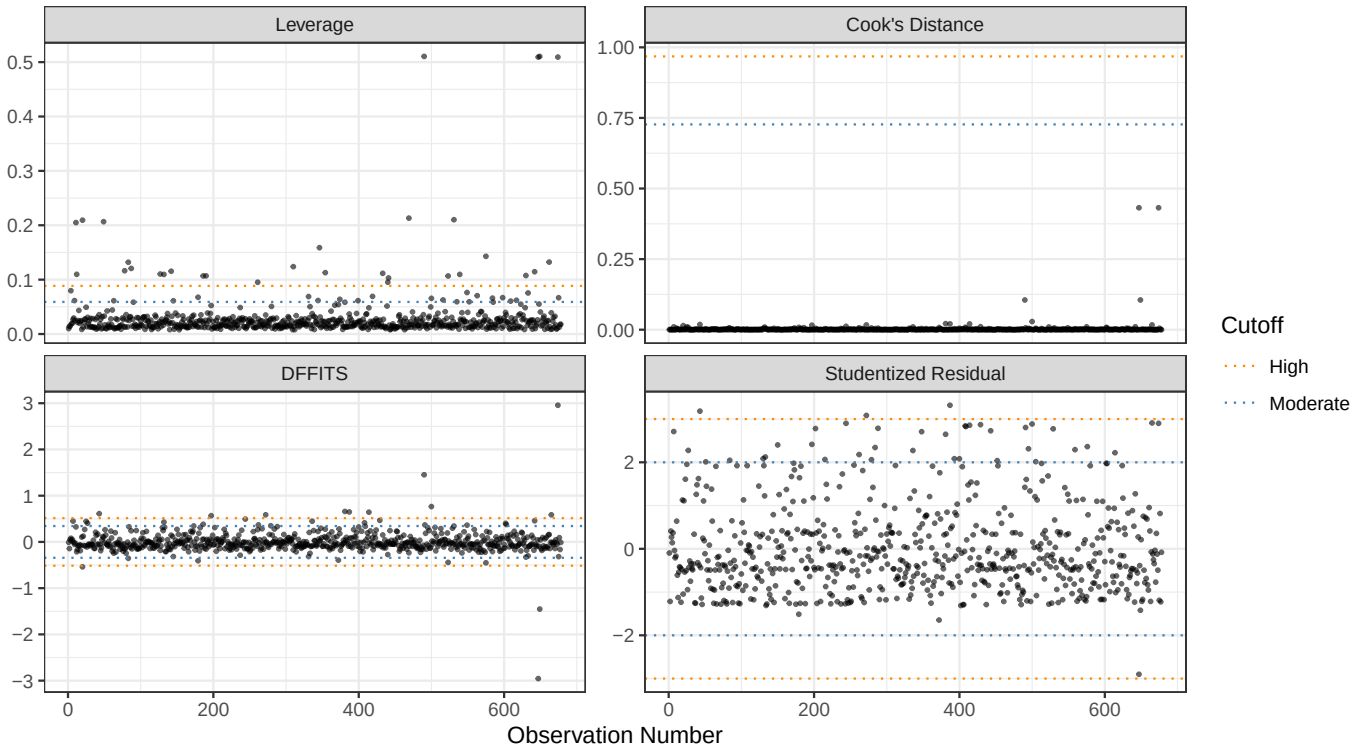



Figure 9: Influence diagnostics for the goal-framing model.




```
augment(m_goal) |> group_by(frame_goal) |> summarize(resid_sd = sd(.resid)) |> kable(digits = 3)
```

| frame_goal | resid_sd |
|---|---|
| Negative | 1.069 |
| Positive | 1.393 |

```r
vif(lm(resp_goal ~ gender + ethnicity + politics + education + frame_goal + age,
       data = framing))
```

```
               GVIF Df GVIF^(1/(2*Df))
gender     1.068292  2        1.016653
ethnicity  1.154549  5        1.014475
politics   1.176864  3        1.027514
education  1.178615  6        1.013789
frame_goal 1.033384  1        1.016555
age        1.060030  1        1.029578
```

```r
bc <- MASS::boxcox(m_goal, plotit = FALSE); bc$x[which.max(bc$y)]
```

```
[1] -0.2
```

- **Gaussian errors:** This is the most problematic of the three models. Most participants chose 1 or 2 ("surely stop"), so the residuals are right-skewed, and the Q-Q plot bends upward in the upper tail (Figure 8).
- **Constant variance:** The residual spread is larger under the positive frame.
- **Box-Cox:** It suggests $\lambda \approx 0$, i.e., a log transformation. Following the lecture's approach, we refit the model on the log scale as a robustness check below, but keep the untransformed model as the primary analysis because it can be interpreted on the rating scale.
- **Influence and multicollinearity:** The same conclusions as for the attribute model apply. No observation is influential, and all VIFs are close to 1.

**Overall $F$-test and effect size.**

```r
anova(m_goal) |> as.data.frame() |> kable(digits = 3)
```

| | Df | Sum Sq | Mean Sq | F value | Pr(>F) |
|---|---|---|---|---|---|
| gender | 2 | 0.792 | 0.396 | 0.248 | 0.780 |
| ethnicity | 5 | 4.882 | 0.976 | 0.612 | 0.691 |
| politics | 3 | 4.477 | 1.492 | 0.936 | 0.423 |
| education | 6 | 9.897 | 1.650 | 1.034 | 0.402 |
| frame_goal | 1 | 41.488 | 41.488 | 26.011 | 0.000 |
| age | 1 | 3.533 | 3.533 | 2.215 | 0.137 |
| frame_goal:age | 1 | 4.555 | 4.555 | 2.856 | 0.092 |
| Residuals | 658 | 1049.521 | 1.595 | NA | NA |

```r
F_to_omega2(f = k_goal$F, df = k_goal$df1, df_error = k_goal$df2)
```

```
Omega2 (partial) |       95% CI
--------------------------------
0.04             | [0.00, 1.00]

- One-sided CIs: upper bound fixed at [1.00].
```

```r
interpret_omega_squared(F_to_omega2(k_goal$F, k_goal$df1, k_goal$df2)$Omega2_partial, rules = "cohen1992")
```

```
[1] "small"
(Rules: cohen1992)
```

The model is statistically discernible but explains little of the variation ($R^2 = 0.062$, adjusted $R^2 = 0.035$; $F(19, 658) = 2.3$, p = 0.001).

```
term_effects(m_goal)
```

Table 4: Effect size (partial omega-squared) for each term, goal framing.

| Term | Partial omega^2 | Lower | Upper | Size |
|---|---|---|---|---|
| gender | 0.000 | 0.000 | 1 | very small |
| ethnicity | 0.000 | 0.000 | 1 | very small |
| politics | 0.000 | 0.000 | 1 | very small |
| education | 0.000 | 0.000 | 1 | very small |
| frame_goal | 0.036 | 0.016 | 1 | small |
| age | 0.002 | 0.000 | 1 | very small |
| frame_goal:age | 0.003 | 0.000 | 1 | very small |

```
coef_table(m_goal)
```

Table 5: Coefficients for the goal-framing model.

| Term | Estimate | SE | t | p | Std-Coef |
|---|---|---|---|---|---|
| (Intercept) | 2.598 | 0.255 | 10.198 | <0.0001 | -0.147 |
| genderMale | 0.031 | 0.101 | 0.311 | 0.756 | 0.024 |
| genderNon-binary / third gender | 0.099 | 0.411 | 0.241 | 0.8098 | 0.077 |
| ethnicityAsian | 0.014 | 0.180 | 0.079 | 0.9373 | 0.011 |
| ethnicityBlack or African American | -0.124 | 0.189 | -0.658 | 0.5105 | -0.097 |
| ethnicityOther | -0.025 | 0.300 | -0.083 | 0.9337 | -0.019 |
| ethnicityAmerican Indian or Alaska Native | -0.301 | 0.572 | -0.526 | 0.5988 | -0.234 |
| ethnicityNative Hawaiian or Pacific Islander | 0.929 | 0.898 | 1.034 | 0.3014 | 0.722 |
| politicsIndependent | -0.052 | 0.120 | -0.431 | 0.6664 | -0.040 |
| politicsRepublican | -0.001 | 0.128 | -0.006 | 0.9953 | -0.001 |
| politicsSomething else | 0.447 | 0.268 | 1.668 | 0.0957 | 0.348 |
| educationLess than high school | 0.221 | 0.904 | 0.244 | 0.8072 | 0.172 |
| educationHigh school graduate | -0.259 | 0.178 | -1.454 | 0.1463 | -0.201 |
| educationSome college | -0.058 | 0.140 | -0.415 | 0.6785 | -0.045 |
| education2 year degree | -0.179 | 0.174 | -1.027 | 0.3049 | -0.139 |
| educationProfessional degree | -0.137 | 0.137 | -1.005 | 0.3155 | -0.107 |
| educationDoctorate | -0.084 | 0.408 | -0.205 | 0.8379 | -0.065 |
| frame_goalPositive | -0.042 | 0.338 | -0.124 | 0.9011 | 0.393 |
| age | -0.013 | 0.006 | -2.246 | 0.025 | -0.123 |
| frame_goalPositive:age | 0.013 | 0.008 | 1.690 | 0.0915 | 0.129 |

**Post hoc testing.**

```
emt_goal <- emtrends(m_goal, specs = "frame_goal", var = "age")
test(emt_goal) |> as.data.frame() |> kable(digits = 4)
```

| frame_goal | age.trend | SE | df | t.ratio | p.value |
|---|---|---|---|---|---|
| Negative | -0.0127 | 0.0057 | 658 | -2.2461 | 0.0250 |
| Positive | 0.0007 | 0.0056 | 658 | 0.1228 | 0.9023 |

```r
pairs(emt_goal, reverse = TRUE) |> as.data.frame() |> kable(digits = 4)
```

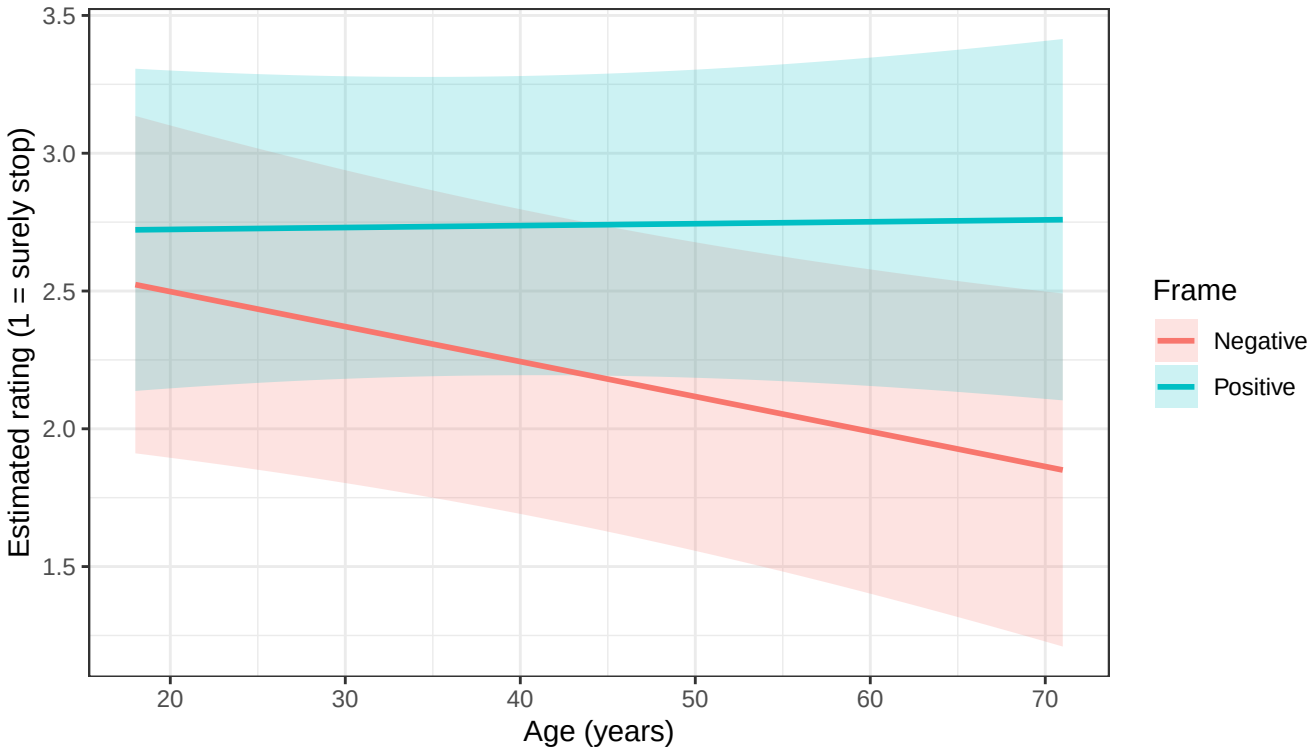
| contrast | estimate | SE | df | t.ratio | p.value |
|---|---|---|---|---|---|
| Positive - Negative | 0.0134 | 0.0079 | 658 | 1.6899 | 0.0915 |

```r
emm_goal <- emmeans(m_goal, specs = "frame_goal", by = "age",
                    at = list(age = c(20, 30, 40, 50, 60, 70)))
pairs(emm_goal, reverse = TRUE) |>
  as.data.frame() |>
  mutate(d = estimate / sd(framing$resp_goal)) |>
  kable(digits = 3)
```

| contrast | age | estimate | SE | df | t.ratio | p.value | d |
|---|---|---|---|---|---|---|---|
| Positive - Negative | 20 | 0.226 | 0.192 | 658 | 1.172 | 0.241 | 0.176 |
| Positive - Negative | 30 | 0.360 | 0.131 | 658 | 2.746 | 0.006 | 0.280 |
| Positive - Negative | 40 | 0.493 | 0.099 | 658 | 4.989 | 0.000 | 0.384 |
| Positive - Negative | 50 | 0.627 | 0.122 | 658 | 5.126 | 0.000 | 0.488 |
| Positive - Negative | 60 | 0.761 | 0.181 | 658 | 4.208 | 0.000 | 0.592 |
| Positive - Negative | 70 | 0.895 | 0.251 | 658 | 3.565 | 0.000 | 0.696 |

```r
emm_plot(m_goal, "frame_goal", "Estimated rating (1 = surely stop)")
```

Figure 10: Estimated marginal mean likelihood of continuing to drink cocoa by frame across age (95% CI).




```r
m_goal_jn <- lm(resp_goal ~ gender + ethnicity + politics + education + pos * age,
                data = framing |> mutate(pos = as.numeric(frame_goal == "Positive")))
jn_goal <- johnson_neyman(m_goal_jn, pred = pos, modx = age, control.fdr = TRUE)
jn_goal
```

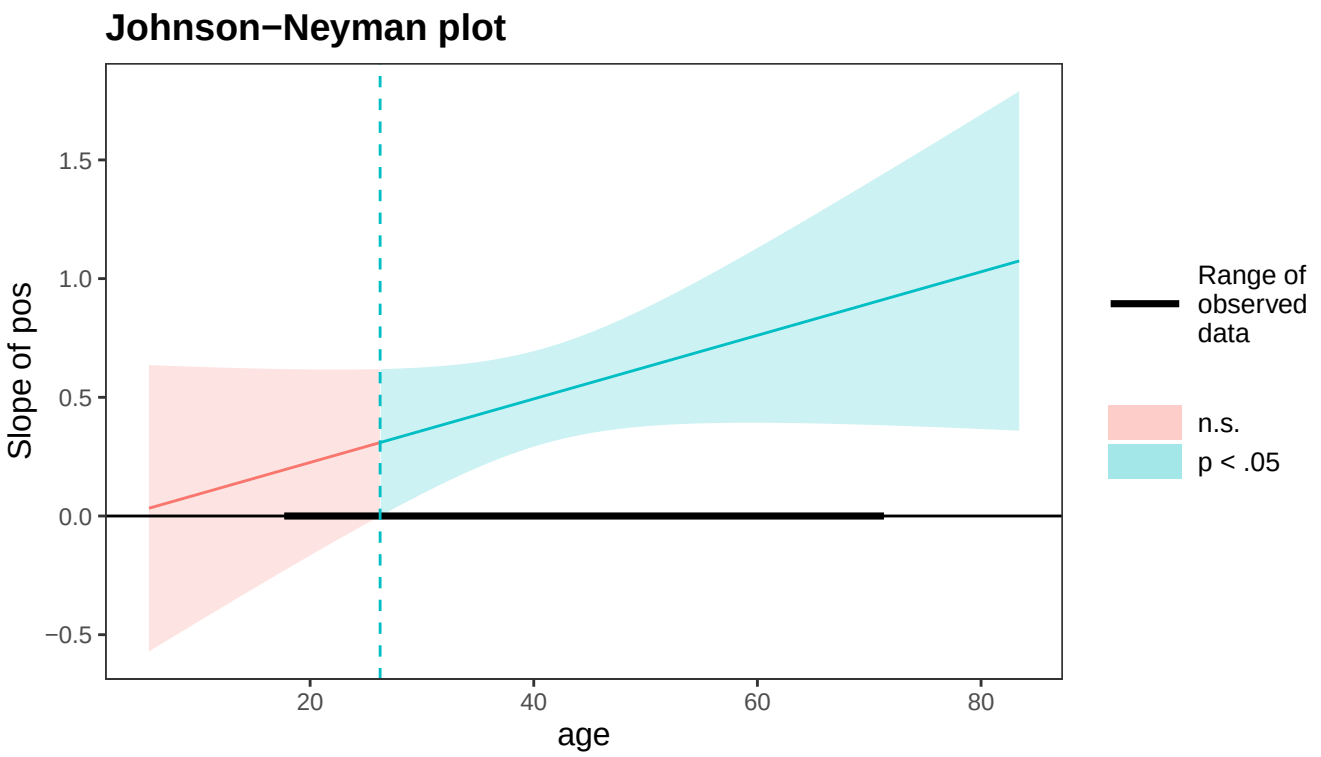
```
JOHNSON-NEYMAN INTERVAL

When age is INSIDE the interval [26.26, 222.81], the slope of pos is p <
.05.

Note: The range of observed values of age is [18.00, 71.00]

Interval calculated using false discovery rate adjusted t = 2.04
```

Figure 11: Johnson-Neyman plot: the framing effect (positive minus negative) across age, goal framing.




**Summary.**

- **Change with age:** The interaction is in the predicted direction ($\hat{\beta}_3 = 0.0134$), but it is not statistically discernible at $\alpha = 0.05$ (p = 0.092), and its effect size is very small.
- **Direction:** The pattern is driven by the negative frame. Under "continuing increases cancer risk," older participants were more likely to say they would stop drinking cocoa (slope = -0.127 per decade, p = 0.025). Under the positive frame, age had essentially no relationship with the rating (0.007 per decade, p = 0.902).
- **Onset:** The Johnson-Neyman analysis shows that the framing effect is not discernible among the youngest adults. It becomes statistically discernible at about age 222.8 and stays discernible through age 71 (Figure 11).

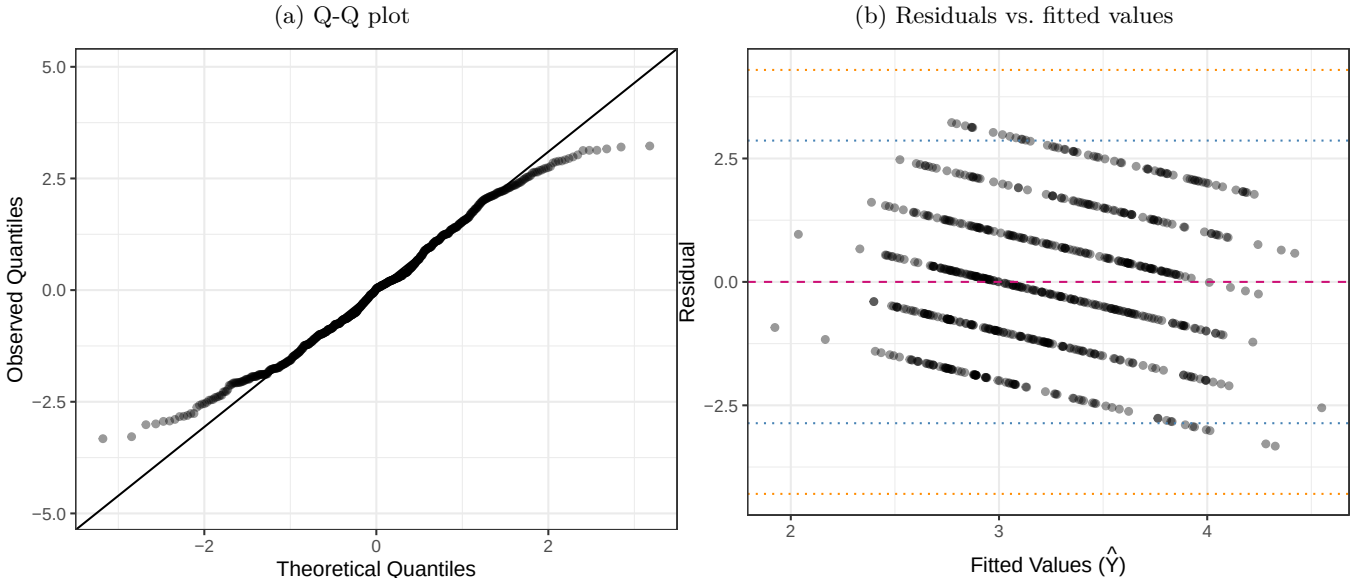
## 3.3 Preference for Chance Choice

```r
m_risky <- lm(resp_risky ~ gender + ethnicity + politics + education +
                frame_risky * age,
              data = framing)
k_risky <- key_stats(m_risky, "frame_risky")
```

**Conditions.**

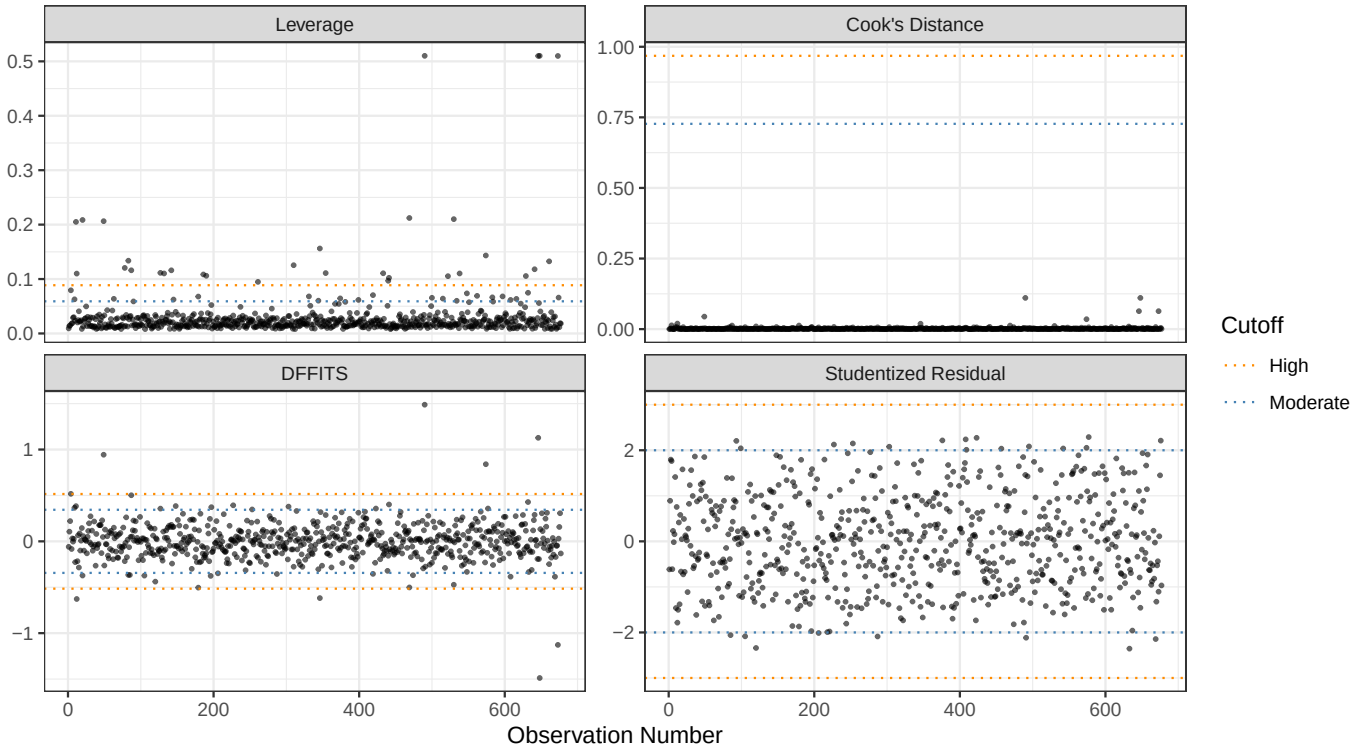
```
r <- resid_plots(m_risky); r$qq; r$rf
```

Figure 12: Residual diagnostics for the risky-choice model.




```
influence_plots(m_risky)
```

Figure 13: Influence diagnostics for the risky-choice model.




```
augment(m_risky) |> group_by(frame_risky) |> summarize(resid_sd = sd(.resid)) |> kable(digits = 3)
```

| frame_risky | resid_sd |
|---|---|
| Negative | 1.471 |
| Positive | 1.351 |

```
vif(lm(resp_risky ~ gender + ethnicity + politics + education + frame_risky + age,
       data = framing))
```

```
                GVIF Df GVIF^(1/(2*Df))
gender      1.065405  2        1.015965
ethnicity   1.150583  5        1.014126
politics    1.184665  3        1.028646
education   1.158332  6        1.012324
frame_risky 1.026221  1        1.013026
age         1.068738  1        1.033798
```

```
bc <- MASS::boxcox(m_risky, plotit = FALSE); bc$x[which.max(bc$y)]
```

```
[1] 0.6
```

- **Gaussian errors:** This model meets the conditions best. Ratings use the whole 1–6 scale, so the residuals are roughly symmetric. The tails are slightly lighter than Gaussian because the scale is bounded (Figure 12).
- **Constant variance:** The residual spread is similar under the two frames.
- **Box-Cox:** $\lambda$ is between 0.5 and 1, so no transformation is needed.
- **Influence and multicollinearity:** There are no influential points, and all VIFs are close to 1. The participant who skipped this question is excluded, giving n = 677.

**Overall *F*-test and effect size.**

```
anova(m_risky) |> as.data.frame() |> kable(digits = 3)
```

| | Df | Sum Sq | Mean Sq | F value | Pr(>F) |
|---|---|---|---|---|---|
| gender | 2 | 1.919 | 0.960 | 0.469 | 0.626 |
| ethnicity | 5 | 12.048 | 2.410 | 1.176 | 0.319 |
| politics | 3 | 32.900 | 10.967 | 5.354 | 0.001 |
| education | 6 | 10.076 | 1.679 | 0.820 | 0.555 |
| frame_risky | 1 | 71.789 | 71.789 | 35.045 | 0.000 |
| age | 1 | 6.386 | 6.386 | 3.118 | 0.078 |
| frame_risky:age | 1 | 10.274 | 10.274 | 5.016 | 0.025 |
| Residuals | 657 | 1345.842 | 2.048 | NA | NA |

```
F_to_omega2(f = k_risky$F, df = k_risky$df1, df_error = k_risky$df2)
```

```
Omega2 (partial) |       95% CI
--------------------------------
0.07             | [0.02, 1.00]

- One-sided CIs: upper bound fixed at [1.00].
```

```
interpret_omega_squared(F_to_omega2(k_risky$F, k_risky$df1, k_risky$df2)$Omega2_partial, rules = "cohen1992
```

```
[1] "small"
(Rules: cohen1992)
```

The model is statistically discernible ($R^2 = 0.097$, adjusted $R^2 = 0.071$; $F(19, 657) = 3.74$, $p < 0.001$).

```r
term_effects(m_risky)
```

Table 6: Effect size (partial omega-squared) for each term, risky-choice framing.

| Term | Partial omega^2 | Lower | Upper | Size |
|---|---|---|---|---|
| gender | 0.000 | 0.000 | 1 | very small |
| ethnicity | 0.001 | 0.000 | 1 | very small |
| politics | 0.019 | 0.003 | 1 | very small |
| education | 0.000 | 0.000 | 1 | very small |
| frame_risky | 0.048 | 0.025 | 1 | small |
| age | 0.003 | 0.000 | 1 | very small |
| frame_risky:age | 0.006 | 0.000 | 1 | very small |

```r
coef_table(m_risky)
```

Table 7: Coefficients for the risky-choice model.

| Term | Estimate | SE | t | p | Std-Coef |
|---|---|---|---|---|---|
| (Intercept) | 2.646 | 0.296 | 8.947 | <0.0001 | 0.093 |
| genderMale | 0.136 | 0.115 | 1.190 | 0.2345 | 0.092 |
| genderNon-binary / third gender | -0.111 | 0.464 | -0.240 | 0.8104 | -0.075 |
| ethnicityAsian | 0.177 | 0.205 | 0.863 | 0.3884 | 0.119 |
| ethnicityBlack or African American | 0.315 | 0.213 | 1.478 | 0.1398 | 0.212 |
| ethnicityOther | -0.048 | 0.340 | -0.143 | 0.8867 | -0.033 |
| ethnicityAmerican Indian or Alaska Native | -0.716 | 0.648 | -1.105 | 0.2696 | -0.482 |
| ethnicityNative Hawaiian or Pacific Islander | 0.180 | 1.017 | 0.177 | 0.8596 | 0.121 |
| politicsIndependent | 0.489 | 0.137 | 3.564 | 4e-04 | 0.329 |
| politicsRepublican | 0.175 | 0.145 | 1.206 | 0.2281 | 0.118 |
| politicsSomething else | 0.089 | 0.304 | 0.294 | 0.7689 | 0.060 |
| educationLess than high school | -0.717 | 1.025 | -0.700 | 0.4843 | -0.483 |
| educationHigh school graduate | -0.099 | 0.202 | -0.490 | 0.6245 | -0.066 |
| educationSome college | -0.009 | 0.159 | -0.058 | 0.9535 | -0.006 |
| education2 year degree | -0.287 | 0.196 | -1.465 | 0.1433 | -0.193 |
| educationProfessional degree | -0.240 | 0.155 | -1.550 | 0.1216 | -0.161 |
| educationDoctorate | -0.349 | 0.463 | -0.753 | 0.4518 | -0.235 |
| frame_riskyPositive | 0.188 | 0.386 | 0.487 | 0.6262 | -0.430 |
| age | 0.018 | 0.006 | 2.824 | 0.0049 | 0.150 |
| frame_riskyPositive:age | -0.020 | 0.009 | -2.240 | 0.0255 | -0.169 |

**Post hoc testing.**

```r
emt_risky <- emtrends(m_risky, specs = "frame_risky", var = "age")
test(emt_risky) |> as.data.frame() |> kable(digits = 4)
```

| frame_risky | age.trend | SE | df | t.ratio | p.value |
|---|---|---|---|---|---|
| Negative | 0.0179 | 0.0063 | 657 | 2.8237 | 0.0049 |
| Positive | -0.0023 | 0.0065 | 657 | -0.3551 | 0.7226 |

```
pairs(emt_risky, reverse = TRUE) |> as.data.frame() |> kable(digits = 4)
```

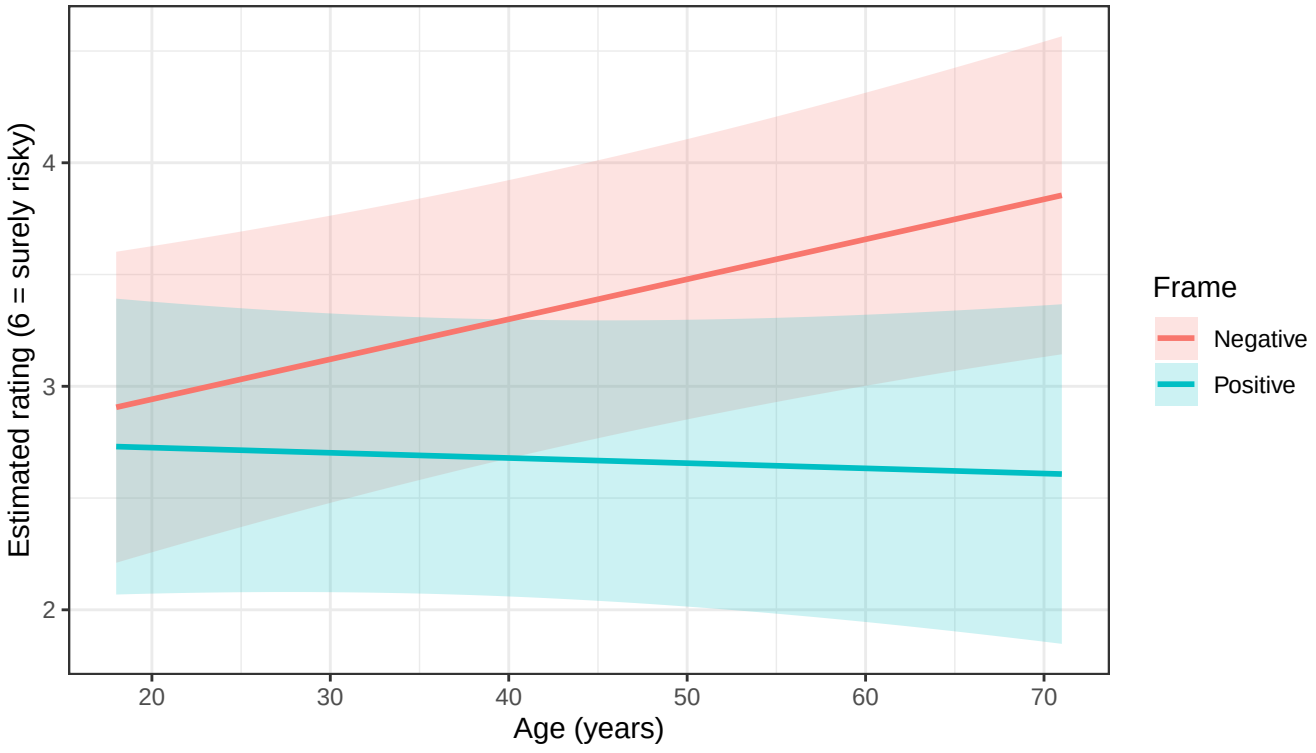
| contrast | estimate | SE | df | t.ratio | p.value |
|---|---|---|---|---|---|
| Positive - Negative | -0.0202 | 0.009 | 657 | -2.2396 | 0.0255 |

```
emm_risky <- emmeans(m_risky, specs = "frame_risky", by = "age",
                     at = list(age = c(20, 30, 40, 50, 60, 70)))
pairs(emm_risky, reverse = TRUE) |>
  as.data.frame() |>
  mutate(d = estimate / sd(framing$resp_risky, na.rm = TRUE)) |>
  kable(digits = 3)
```

| contrast | age | estimate | SE | df | t.ratio | p.value | d |
|---|---|---|---|---|---|---|---|
| Positive - Negative | 20 | -0.216 | 0.219 | 657 | -0.987 | 0.324 | -0.146 |
| Positive - Negative | 30 | -0.418 | 0.149 | 657 | -2.815 | 0.005 | -0.282 |
| Positive - Negative | 40 | -0.621 | 0.112 | 657 | -5.553 | 0.000 | -0.418 |
| Positive - Negative | 50 | -0.823 | 0.138 | 657 | -5.943 | 0.000 | -0.554 |
| Positive - Negative | 60 | -1.025 | 0.205 | 657 | -4.993 | 0.000 | -0.690 |
| Positive - Negative | 70 | -1.227 | 0.285 | 657 | -4.301 | 0.000 | -0.826 |

```
emm_plot(m_risky, "frame_risky", "Estimated rating (6 = surely risky)")
```

Figure 14: Estimated marginal mean preference for the risky option by frame across age (95% CI).



```r
m_risky_jn <- lm(resp_risky ~ gender + ethnicity + politics + education + pos * age,
                 data = framing |> mutate(pos = as.numeric(frame_risky == "Positive")))
jn_risky <- johnson_neyman(m_risky_jn, pred = pos, modx = age, control.fdr = TRUE)
jn_risky
```

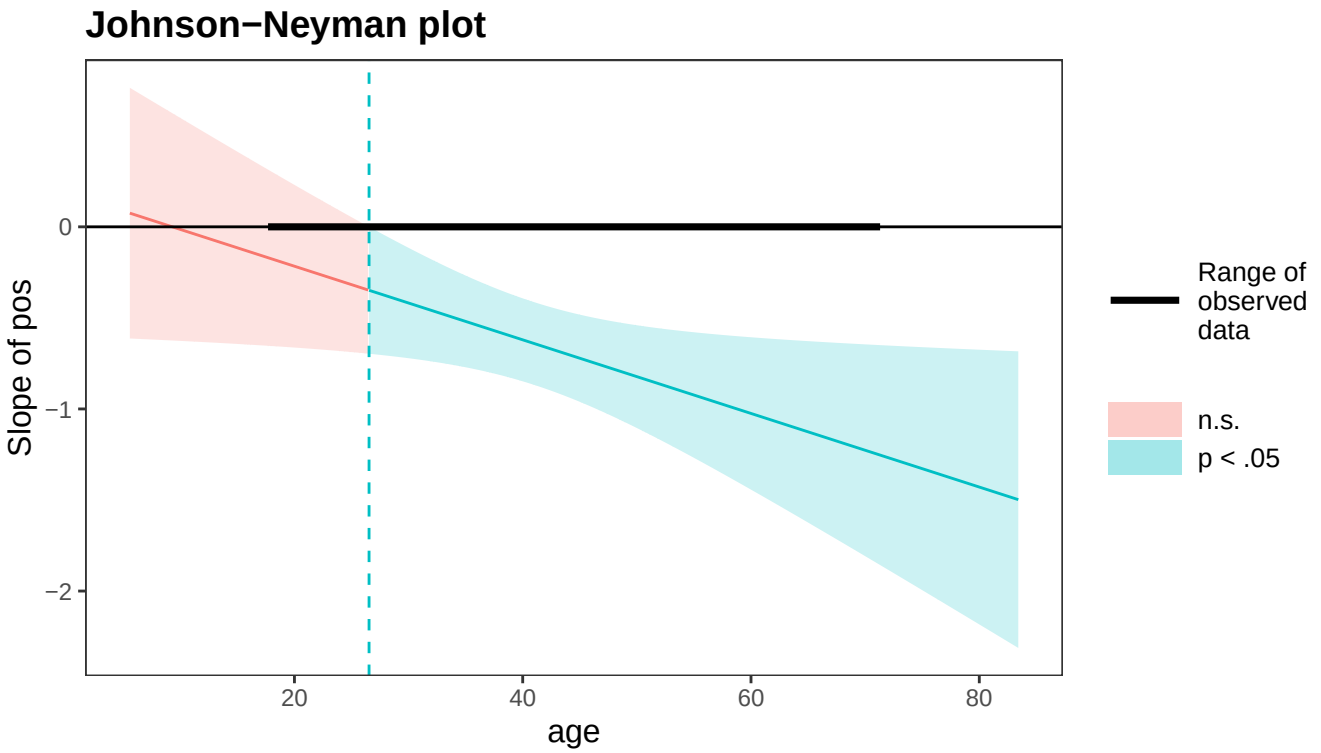
```
JOHNSON-NEYMAN INTERVAL

When age is OUTSIDE the interval [-313.21, 26.54], the slope of pos is p <
.05.

Note: The range of observed values of age is [18.00, 71.00]

Interval calculated using false discovery rate adjusted t = 2.04
```

Figure 15: Johnson-Neyman plot: the framing effect (positive minus negative) across age, risky-choice framing.




**Summary.** Risky choice is the only scenario with a statistically discernible frame × age interaction ($\hat{\beta}_3$ = -0.0202, p = 0.025), although its effect size is very small.

- **Direction:** The change is driven entirely by the negative frame. When the options were described in terms of deaths, preference for the risky program rose with age (0.179 points per decade, p = 0.005). When they were described in terms of lives saved, age had no relationship with preference (-0.023 per decade, p = 0.723).
- **Onset:** The framing effect is not discernible among the youngest participants. It becomes statistically discernible at about age 26.5 and grows from there (Figure 15).

### 3.3.1 Robustness checks

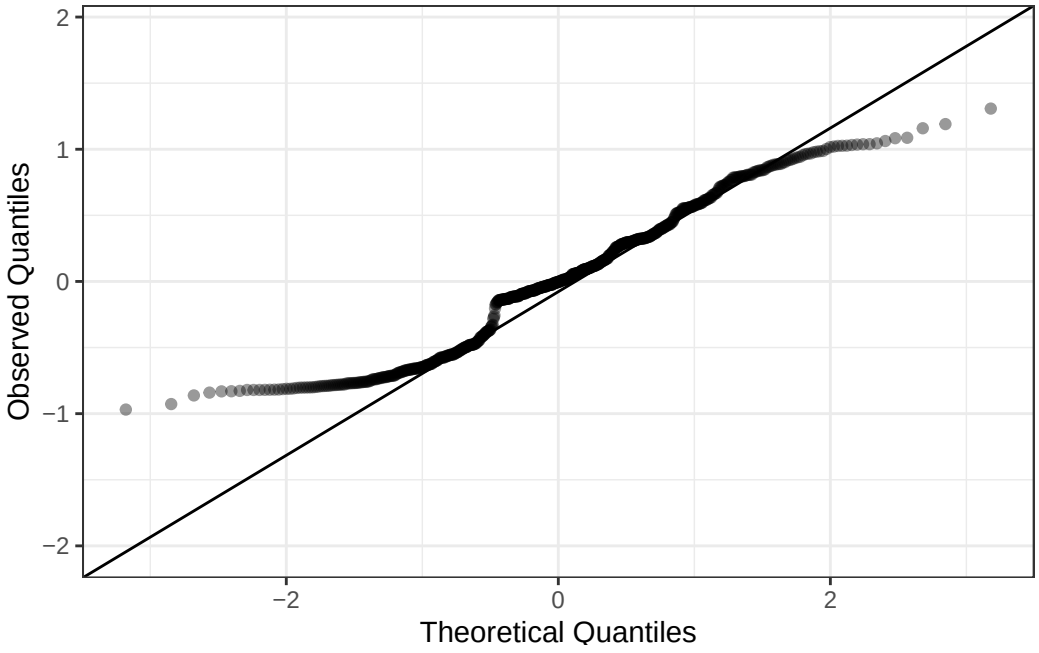
The residual analyses raised two concerns: unequal residual variance between the frames, and right skew in the goal model. We check whether either one changes the conclusions about the interaction.

```r
robust_int <- function(mod, frame) {
  emtrends(mod, specs = frame, var = "age", vcov. = vcovHC(mod, type = "HC3")) |>
    pairs(reverse = TRUE) |>
    as.data.frame()
}
bind_rows(Attribute      = robust_int(m_att,   "frame_attribute"),
          Goal           = robust_int(m_goal,  "frame_goal"),
          `Risky choice` = robust_int(m_risky, "frame_risky"),
          .id = "Scenario") |>
  select(Scenario, estimate, SE, t.ratio, p.value) |>
  kable(digits = 4)
```

Table 8: Frame × age interaction (difference in age slopes, Positive − Negative) with heteroscedasticity-robust (HC3) standard errors.

| Scenario | estimate | SE | t.ratio | p.value |
|---|---|---|---|---|
| Attribute | 0.0075 | 0.0043 | 1.7583 | 0.0792 |
| Goal | 0.0134 | 0.0075 | 1.7784 | 0.0758 |
| Risky choice | -0.0202 | 0.0095 | -2.1174 | 0.0346 |

```r
m_goal_log <- lm(log(resp_goal) ~ gender + ethnicity + politics + education +
                   frame_goal * age,
                 data = framing)
r <- resid_plots(m_goal_log); r$qq
```




```r
tidy(m_goal_log) |> filter(str_detect(term, "frame_goal|^age")) |> kable(digits = 4)
```

| term | estimate | std.error | statistic | p.value |
|---|---|---|---|---|
| frame_goalPositive | -0.0654 | 0.1426 | -0.4589 | 0.6465 |
| age | -0.0061 | 0.0024 | -2.5674 | 0.0105 |
| frame_goalPositive:age | 0.0066 | 0.0033 | 1.9716 | 0.0491 |

- **Robust standard errors:** Using heteroscedasticity-robust (HC3) standard errors leaves every conclusion unchanged (Table 8). The attribute and goal interactions remain not discernible, and the risky-choice interaction remains discernible.
- **Log-transformed goal model:** Log-transforming the goal response removes most of the skew. The interaction then sits right at the $\alpha = 0.05$ boundary, with the same direction and pattern: older adults are more likely to stop under the negative frame only. The goal-framing result is borderline, but the qualitative pattern does not depend on how we model it.

# 4 Discussion

# References

Xu, Lance. 2024. "Advanced Age Predicts Increased Susceptibility to Attribute, Goal, and Risky-Choice Framing in Negative Frame Valences." *Humanities and Social Sciences Communications* 11 (1): 217.